

LESSON 9.9

SCALE FACTORS OF DILATIONS



LESSON INTRODUCTION

MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

LEARNING TARGETS

- I can determine the scale factor of a given dilation.
- I can solve real-world problems that involve similar triangles.

BENCHMARK COHERENCE

BUILDING ON

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

WORKING TOWARD

MA.9.12.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

ESSENTIAL QUESTIONS

- What is the scale factor?
- How did you determine the scale factor?
- How can you use scale factor to solve real-world problems?

LESSON NARRATIVE

Students discover that the scale factor can be found by: (1) dividing the corresponding side length of the image by the corresponding side length of the preimage; (2) dividing the distance from the center to vertex on the image by the distance from the center to corresponding vertex on the preimage. Students analyze samples of student work and find scale factors in real-world contexts.

COMPONENT SUMMARY

9.9.1 WARM-UP (~5 MIN)

Compare dilations

9.9.3 COLLABORATION (15 MIN)

Analyze errors in dilations

9.9.5 WRAP-UP (~5 MIN)

Determine a scale factor

9.9.2 EXPLORATION (20 MIN)

Identify scale factors of dilations

9.9.4 GUIDED PRACTICE (15 MIN)

Identify scale factor and coordinates in real-world dilations

RESOURCES

GLOSSARY

Key Terms:

- Scale factor

Additional Terms: N/A

MATERIALS

- (Optional) Highlighters or Colored Pencils
- (Optional) Graph Paper
- (Optional) Scissors

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may find the reciprocal of the scale factor.
- Students may not use corresponding side lengths when determining scale factor.

THINGS TO CONSIDER

- Have students predict whether a scale factor will be greater than or less than 1. Doing so will help them consider if the scale factor they calculate is reasonable.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 9.9.2 Exploration ask students what they notice and wonder about the distances from the vertices to the centers of dilations and side lengths of the preimages and images. Focus student observations to help them discover the ways to determine scale factors.

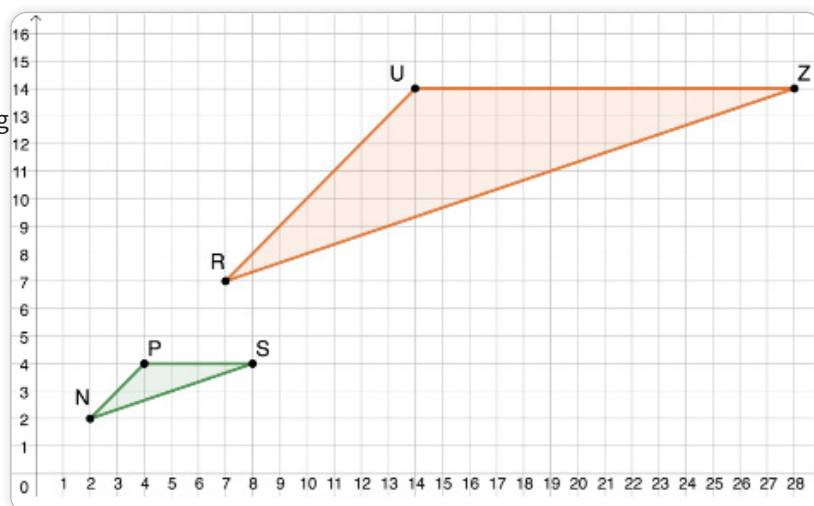
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 9.9.3 Collaboration students identify and correct errors in samples of student work determining scale factors. After describing and discussing the errors, students will be less likely to make the errors themselves.

DIFFERENTIATE INSTRUCTION

Provide a visual entry-point by having students use different color highlighters or colored pencils to color code corresponding sides and vertices in 9.9.4 Guided Practice. Provide a kinesthetic entry-point by prompting students to cut out or draw the preimages and images on graph paper in 9.9.2 Exploration. The discussions during 9.9.3 Collaboration provide an auditory entry-point.



9.9.1 WARM-UP: BELL WORK

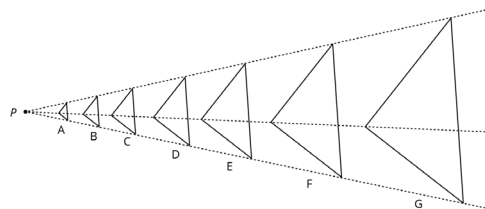
ENRICHMENT

- How can you tell if the scale factor of a dilation is between 0 and 1?
- How can you tell if the scale factor of a dilation is greater than 1?
- What does the image of a figure look like if the scale factor is 1?



9.9.1 Warm-Up: Bell Work

1. All of the triangles shown are dilations of Triangle D . The dilations use the same center, point P , but different scale factors.



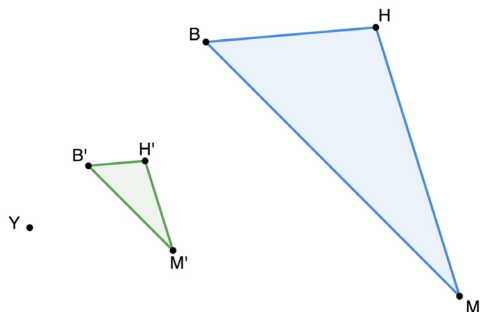
- a. How do Triangles A , B , and C compare to the preimage?
Triangles A , B , and C are all smaller than the preimage.
- b. How do Triangles E , F , and G compare to the preimage?
Triangles E , F , and G are all bigger than the preimage.
- c. What do you know about the different scale factors used based on your answers to Parts A and B?
The scale factors for triangles A , B , and C are all between 0 and 1, and the scale factors for triangles E , F , and G are all greater than 1.



9.9.2 Exploration: Identifying the Scale Factor of Dilations

Work with your partner to complete the following.

- Triangle BHM is dilated using point Y as the center, as shown.



- Complete the statement.

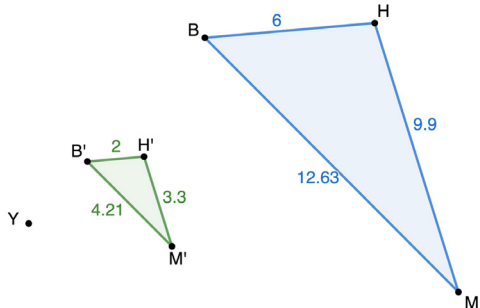
The scale factor of this dilation is

- ☐ between 0 and 1.
- ☐ equal to 1.
- ☐ greater than 1.

- Explain your reasoning.

I know the scale factor is between 0 and 1 because the image is smaller than the preimage.

- The measurements, in units, of the side lengths of each triangle have been added to the figure.



Myrical said the scale factor used is $\frac{1}{3}$, while Joshua said it is 3. Explain who is correct.

Myrical is correct because the scale factor needs to be between 0 and 1 because the image is smaller than the preimage.

- Discuss with your partner how the distance between points Y and M should compare to the distance between points Y and M' . Then, complete the statement.

The distance between points Y and M should be 3 times the distance between Y and M' .

9.9.2 EXPLORATION: IDENTIFYING THE SCALE FACTOR OF DILATIONS

DIFFERENTIATE INSTRUCTION

Point out that throughout the lesson, the preimages are shown in green and the transformed images are shown in blue. Provide a kinesthetic entry-point by prompting students to cut out or draw the preimages and images on graph paper.

ENRICHMENT

- What is the center of dilation of the figures?
- How far is each vertex from the center?
- What are the lengths of the sides of the triangles?

TEACHER MOVES

Facilitate a discussion prompting students to make connections between scale factor and their work in Unit 6 with similar triangles and finding the constant of proportionality.

9.9.2 EXPLORATION: IDENTIFYING THE SCALE FACTOR OF DILATIONS (CONTINUED)

TEACHER MOVES

Notice and Wonder

Ask students what they notice and wonder about the distances from the vertices to the centers of dilations and side lengths of the preimages and images. Focus student observations to help them discover the ways to determine scale factors.

ENRICHMENT

- When would it be better to use the side lengths?
- When would it be better to use the distance from the center to the vertex?

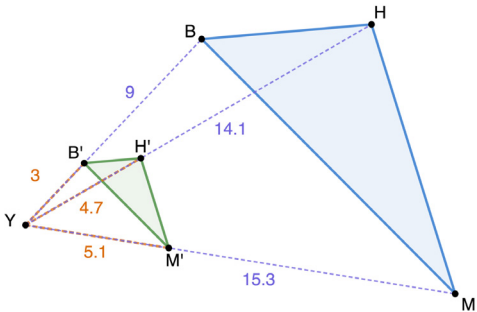
DIFFERENTIATE INSTRUCTION

Consider projecting the images so students can see them more clearly. Prompt students to color code the corresponding vertices and sides with highlighters or colored pencils.

QUESTIONING

Challenge students to create a two-step transformation that includes a dilation and either a rotation, reflection, or translation.

- e. The distances, in units, between the center of dilation and each vertex on the preimage are shown in purple. The distances, in units, between the center of dilation and each vertex on the image are shown in orange.



Explain whether the information given confirms your answer to Part D.
This information confirms my answer because the distance between points Y and M is 15.3 and $5.1 \times 3 = 15.3$.

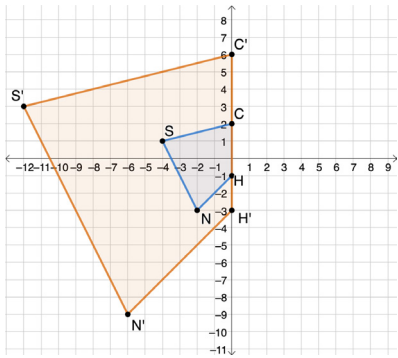


The scale factor of a dilation can be determined using the lengths of the corresponding sides of the preimage or using the distances between the center of dilation and the vertices of the preimage and image.

$$\text{Scale factor} = \frac{\text{corresponding side length of image}}{\text{corresponding side length of pre-image}}$$

$$\text{Scale factor} = \frac{\text{distance from center to vertex on the image}}{\text{distance from center to corresponding pre-image vertex}}$$

2. Quadrilateral *CHNS* is dilated using the origin as the center, as shown.



- a. Complete the table.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point <i>C</i>	(0, 2)	Point <i>C'</i>	(0, 6)
Point <i>H</i>	(0, -1)	Point <i>H'</i>	(0, -3)
Point <i>N</i>	(-2, -3)	Point <i>N'</i>	(-6, -9)
Point <i>S</i>	(-4, 1)	Point <i>S'</i>	(-12, 3)

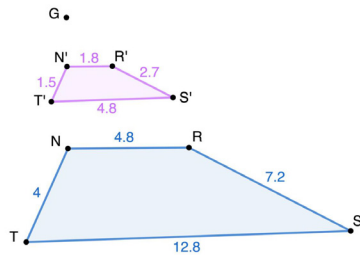
- b. Describe the relationship between the coordinates of the vertices on the preimage and the corresponding coordinates on the image.
The coordinates of the vertices of the image are 3 times the coordinates of the corresponding vertices of the preimage.
- c. What is the scale factor of the dilation?
 3



9.9.3 Collaboration: Error Analysis

Work with your partner to complete the following.

1. Quadrilateral $NRST$ is dilated using point G as the center of dilation.



Latonya found the scale factor of dilation to be $\frac{8}{3}$. Her work is shown.

$$\text{Scale Factor} = \frac{12.8}{4.8} = \frac{8}{3}$$

- a. Is Latonya correct?

No

- b. Explain your reasoning.

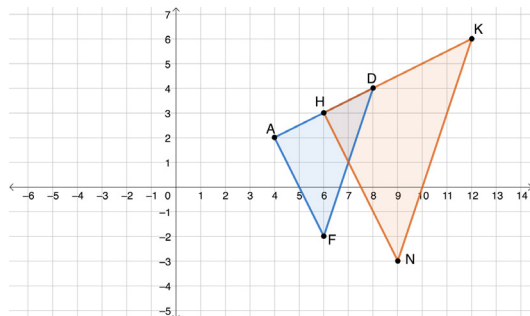
Latonya is incorrect because she found the reciprocal of the scale factor. When finding the scale factor of dilation, you find the ratio $\frac{\text{image}}{\text{preimage}}$, so the scale factor should be $\frac{3}{8}$.

9.9.3 COLLABORATION: ERROR ANALYSIS

ENRICHMENT

- How is scale factor calculated?
- What will occur if a scale factor is greater than 1?
- What will occur if a scale factor is less than 1?
- Is the image larger or smaller than the preimage?
- Is the scale factor reasonable?

2. On the coordinate grid, ΔHKN is the result of dilating ΔADF using the origin as the center of dilation.



Martell says, "The scale factor of this dilation is 3. To find the scale factor, I found the coordinates of point A which were $(4, 2)$, and then I found the coordinates of point K which were $(12, 6)$. I know the scale factor is 3 because $4 \times 3 = 12$ and $2 \times 3 = 6$."

- a. Explain the error Martell made.

Martell is incorrect because he was comparing the coordinates of two points that are not corresponding.

- b. What scale factor was used to dilate ΔADF ?

1.5

DIFFERENTIATE INSTRUCTION

The discussions during this collaboration provide an auditory entry-point. Prompt students to describe the errors and how to correct them verbally.

9.9.4 GUIDED PRACTICE: DILATIONS IN THE REAL WORLD

TEACHER MOVES

Share additional information, photos, and videos about the use of LiDAR by archaeologists.

QUESTIONING

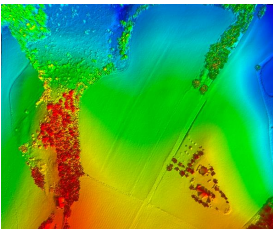
Consider inviting a guest speaker who uses scale factors in a career (i.e. archaeologist, architect, real estate agent, artist) to join the class in-person or virtually. Prompt students to prepare questions to ask the guest speaker.

DIFFERENTIATE INSTRUCTION

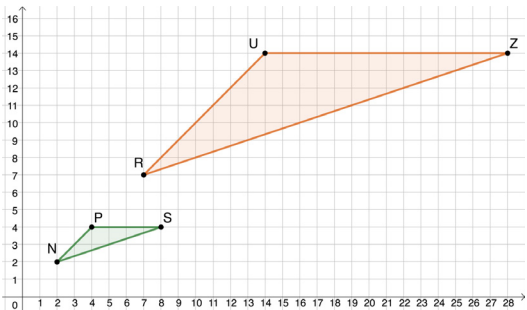
Read questions aloud and discuss words students are unfamiliar with. Encourage students to highlight or underline key words and quantities needed to solve the questions.

9.9.4 Guided Practice: Dilations in the Real World

- Archaeologists have begun using LiDAR (light detection and ranging) technology to map the shape of the ground underneath dense rainforests and jungles. Using this technology, they have discovered numerous ancient cities that have been hidden for thousands of years.



An archaeologist discovered two triangular structures using LiDAR. She overlaid her image on a coordinate grid to analyze the relationship between the structures.



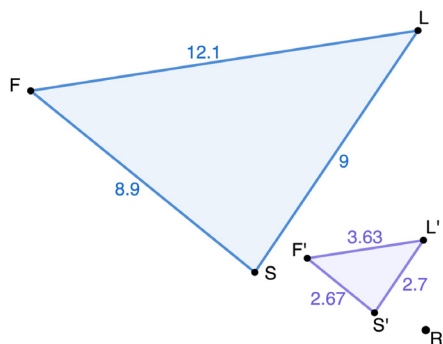
- The archaeologist found that the larger triangle, ΔRUZ , could be obtained by dilating the smaller triangle, ΔNPS , using the origin as the center of dilation. Identify the scale factor of the dilation.
3.5
- Suppose the archaeologist found a third structure that is a dilation of ΔNPS using a scale factor of 11.5 centered at the origin. Determine what the coordinates of the third structure, $\Delta N'P'S'$, would be.

Vertex	Coordinates
N'	(23, 23)
P'	(46, 46)
S'	(92, 46)



9.9.5 Wrap-Up: Ticket Out the Door

- The figure shows the result of dilating $\triangle FLS$ using point R as the center of dilation. Measurements, in units, of each side are given.



Determine the scale factor of the dilation. Show your work.

$$\frac{\text{image}}{\text{pre-image}} = \frac{2.7}{9} = 0.3$$

9.9.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Ask students how they found the scale factor. Prompt them to consider if the scale factor they calculated is reasonable. Finally, inquire how they would have calculated the scale factor if the preimage and image were on the coordinate plane.

ENRICHMENT

Before doing any calculations, do you predict the scale factor to be greater than or less than 1? Why?